

Behavioral motivation

Adaptive and asymmetric reference updating
Same wealth can imply different gain-loss domains

Dynamic-reference CPT model

State: wealth, reference, and portfolio
Dirichlet policy for long-only allocation

Learning method

Unbiased frozen-target gradient estimator
Normalized projected online update

Statistical guarantees

Exact unbiasedness, finite variance, and CLT
Fixed-dimensional minimax trajectory-budget rate

Online-tracking guarantees

Projected residual and dynamic local regret
Stationary-set tracking under an error bound

Empirical validation

E1-E4: mechanism, estimator diagnostics, online tracking, and ablation / robustness
E5-E7: chronological A-share replay, held-out investor behavior, and heterogeneous-agent interaction